

Project Title:

Triplog

Project summary: (max. 500 words) (Please state the main objectives of the project and its realistic constraints clearly, and the broadly final results, also ensure that the project is attainable in one semester)

Travelers today face a common frustration, planning and managing a trip requires juggling multiple applications at once. One app tracks expenses, another visualizes routes, and yet another coordinates group plans — but none of them talk to each other. This fragmentation forces users to manually piece together their travel data, making the overall experience unnecessarily complex and error-prone.

This project proposes a single platform where every aspect of a trip — from setting and managing budget and route planning to group coordination and looking back on the journey — is handled in one place.

A traveler using this service can set a travel budget before departure, scan receipts on the go to automatically log expenses, and visualize their journey across an interactive map in real time. When traveling with friends, the entire group shares the same live view of spending and routes, making collective decisions easier and more transparent. Once the trip ends, the platform generates a personalized travel report that summarizes the journey and suggests future destinations based on the user's travel style.

The project has several realistic constraints. OCR-based receipt scanning will be optimized for printed receipts and may have limitations with handwritten or low-quality images. The 3D route visualization will support a defined set of map data, and AI-generated reports will depend on the volume and quality of data accumulated during the trip. Real-time collaboration features will be scoped to core group functions to ensure stability within the development timeline.

The final deliverable is a working web application prototype that demonstrates all four core features: expense management, route visualization, group collaboration, and

AI-generated travel reporting. The workload is distributed across four team members with clearly defined roles in AI/system design, backend development, and UI implementation, and the full prototype is realistically achievable within a single semester.

Problem Statement: (max. 200 words) (Define what is the problem the project will try to solve)

The idea for this project originated from a common experience shared among friends during group trips. When traveling together, managing expenses, coordinating plans, and keeping track of shared information requires constant back-and-forth across multiple apps — splitting bills on one app, checking routes on another, and sharing plans over messaging. This everyday frustration highlighted a clear gap, there is no single platform designed to support the full experience of traveling together.

Existing applications each address only isolated aspects of travel — some focus on route planning, others on expense recording, and others on group coordination — but none unify these into a seamless workflow. Users must manually enter every expense with no automation, route and spending data exist separately with no way to connect them, and when one group member updates a plan, others cannot see it in real time. Once a trip ends, there is no tool that transforms accumulated travel data into personalized insights or future recommendations.

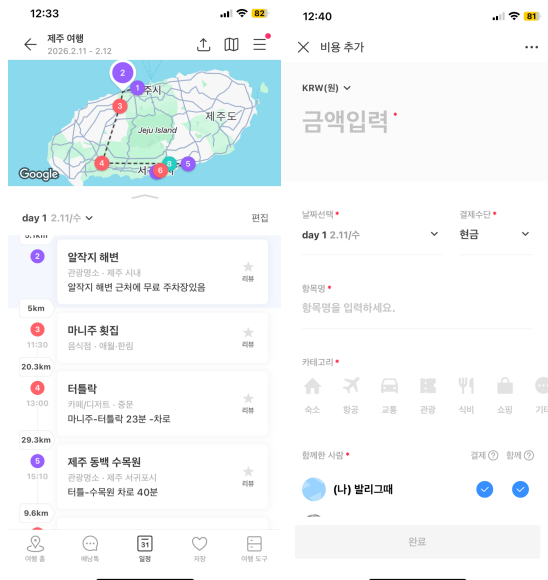
These gaps create four core points: inefficient expense tracking, no integrated route-expense analysis, limited group collaboration support, and no post-travel insight

generation. This project aims to build a unified platform that connects every stage of the travel lifecycle.

Related works (No words limit) (describe what are the existing solutions available to solve this problem and you can include Figures or Tables to this section).

Current travel applications provide a variety of functions, including travel planning, expense tracking, and route visualization. However, these functions are typically offered separately and are not integrated within a single platform. As a result, users are required to use multiple applications to manage different aspects of their travel. This leads to a fundamental limitation in which travel-related data is fragmented rather than managed in a unified manner.

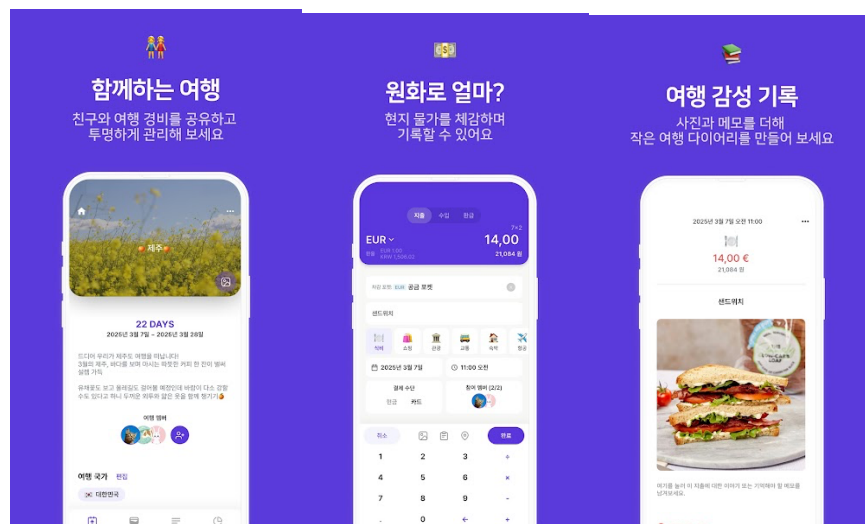
[Travel Route Perspective]



Triple is a representative travel application that focuses on travel planning and reservation systems. By integrating map-based information, it enables users to visualize travel routes and access detailed information about destinations along with booking functionalities.

Although it includes an expense management feature, it relies on manual input, requiring users to enter items and amounts themselves. This limits usability and reduces overall convenience in managing travel expenses.

[Travel Budget Perspective]



TrabeePocket is an application specialized in travel expense management, allowing users to record expenses by category and review spending summaries over a travel period. It also provides additional features such as currency conversion and diary-based recording.

These features effectively support individual expense tracking; however, the application is primarily designed for personal use and offers limited collaboration functionality through a view-only sharing structure. Consequently, it does not support shared expense management or integration with other travel-related data such as routes.

[Integrated Travel Service]

The "Travel Integration Service utilizing Generative AI and Amadeus API" was developed to address the complexity of travel planning by integrating multiple functionalities into a single platform. It provides flight and accommodation booking through the Amadeus API and generates personalized travel itineraries using GPT-based generative AI. Additionally, it includes OCR-based expense tracking and integrates various travel-related information such as weather and exchange rates through multiple Open APIs.

These features highlight the potential of integrating diverse travel functionalities within a unified system and demonstrate the effectiveness of automation techniques such as OCR in reducing user input effort.

However, despite these advantages, the system remains primarily focused on travel planning and information aggregation. It lacks deeper integration between different types of travel data, such as expenses and routes, and does not sufficiently support collaborative features for group-based travel management.

Despite these advancements, existing services still fail to integrate the entire travel process into a continuous and connected workflow.

To address this limitation, this project proposes a unified platform that integrates travel expenses, routes, and collaboration data into a single system. In particular, beyond simple feature aggregation, the system emphasizes user collaboration and shared experiences, enabling users to collectively plan, manage, and reflect on their travel.

Proposed Solution (No words limit) (describe the team vision to solve this project problem. This section can include any technical details. Please note that the described solution in this section can be changed later once you start implementing the project solution) You can elaborate on your idea by adding Figures, resources, references, or extra explanations to clarify the proposed solution

The primary objective of this project is to provide a ***unified platform*** for managing travel spending and routes, focusing on the following core functionalities:

1. **Intelligent Spending and Budget Management System**

This system offers integrated management of fragmented travel expenses and provides actionable insights through budget overrun risk analysis. By analyzing spending patterns against the initial budget, it provides real-time alerts and predictive insights to prevent overspending.

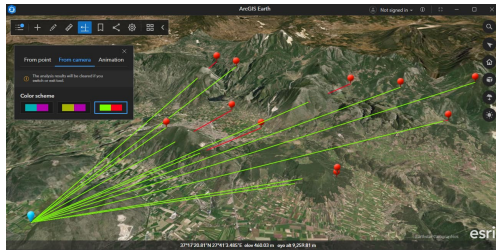
- AI Smart Receipt: Extracting raw text from receipt images via **Google Cloud Vision OCR** and processing raw text into structured JSON data by LLM-based parser.

2. **Immersive Route Visualization and Integrated Reporting**

The platform supports collaborative route planning and provides a 3D interface

for immersive route visualization. And also, the system synthesizes spending and route data to generate AI-driven reports that recommend future travel destinations based on the user's travel theme, ensuring seamless management across the entire travel lifecycle.

- 3D Interface: it will give an immersive geospatial interface using **CesiumJS** and 3D Tiles to render high-fidelity travel trajectories



- AI reporting: Utilizing a RAG (Retrieval-Augmented Generation) architecture to fetch trip data and synthesize it through a LLM to generate personalized reports.

3. Collaborative Social Features

By incorporating sharing features such as Invite Codes, the platform allows multiple users to co-create travel plans. Friends can participate in the planning stage together and access shared real-time data on spending and routes, fostering a highly collaborative travel experience.

Main functions of the project (Items that describe the main functions of the project. These functions can be modified or increased later in the implementation stage)

The proposed system aims to integrate fragmented travel data, including expenses, routes, and collaboration, into a <i>unified platform</i> . To achieve this, the main functions of the system are structured into four key categories: budget and expense management, travel route visualization, collaboration, and travel experience insights.		
Category	Objective	Functions
Budget & Expense Management	To integrate and manage fragmented travel expenses and support data-driven financial decisions	- Budget Planning: Set travel budget in advance and use it as baseline for analysis - AI Smart Receipt (OCR): Extract expense data from receipt images and automatically categorize spending - Currency Conversion: Convert foreign currency into KRW using real-time exchange rates - Spending Risk Analysis: Analyze spending pace vs remaining budget and notify overspending risk
Travel Route & Visualization	To provide intuitive and interactive visualization of travel routes	- Route Visualization: Display travel routes using Google Maps API - 3D Route Rendering: Enhance visualization using CesiumJS and 3D Tiles
Collaboration	To enable shared travel	- Shared Expense

	management and enhance group interaction during trips	Management: Distinguish between individual and group expenses and manage participants - Collaborative Decision Making: Vote-based destination selection among group members - Real-time Interaction: WebSocket-based real-time updates for group activities
Travel Experience & Insights	To transform travel data into meaningful insights and post-travel experiences	- LLM-based Travel Report: Generate structured reports based on expense and route data - Travel Style Analysis: Analyze user travel patterns and preferences - Next Destination Recommendation: Suggest future travel destinations based on past travel data

Time Plan and Roles of Team Members (this includes time plan for the project stages)

Number of Students (min. 3 students, max. 4 students):

List of students:

Student ID	Student Name	Contact or Email	Initial Project Role
2023312897	Seoeun Park (박서은)	pse0420@g.skku.edu	AI model integration, system design, database design
2022311204	Jeongwoo Jang (장정우)	ndn1020@g.skku.edu	Backend development
2022312601	Jungkyo In (인정교)	dreamtree333@g.skku.edu	Frontend development, data visualization UI
2018311923	Haein Park (박혜인)	phi000215@gmail.com	Frontend development, UI design

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